

POLYTECHNIQUE MONTRÉAL

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Waste Twin: A Digital Twin of a Smart Waste Collection System-of-Systems

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LOG6953FE Project Report

Avril 2025

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1 INTRODUCTION

In an era where cities are undergoing digital transformation to become more intelligent, efficient, and responsive, the application of emerging technologies such as the Internet of Things (IoT) and Digital Twins (DTs) has become central to addressing complex urban challenges. Among these, municipal solid waste management remains a critical area requiring innovation. Inefficient collection cycles, environmental degradation, increased waste generation, and operational disruptions due to climate events for example have made traditional waste collection systems insufficiently resilient and unsustainable in many urban areas.

In Montreal, for instance, public dissatisfaction has grown over recurring service delays caused by inclement weather and reduced pick-up frequencies, as well as concerns about the health and environmental risks posed by overflowing garbage bins and poorly managed waste zones¹². At the same time, political debates continue to intensify regarding landfill expansion and the reception of hazardous waste imports³⁴. These concerns underscore the pressing need for more adaptive, real-time, and data-driven approaches to waste collection and resource management in smart cities.

This report introduces **WasteTwin**, an ongoing project of the realisation of a Smart Waste Collection System-of-Systems (SoS) using IoT components and its DT to simulate, monitor, and optimize waste collection operations in an urban context. The project demonstrates how digital twinning can be used to mirror the physical state of a city’s waste collection system in real time, and later-on supporting decision-making and enhancing the SoS resilience—defined as the ability of a system to anticipate, absorb, and adapt to disruptions while maintaining functionality (Hosseini, Barker, & Ramirez-Marquez, 2016).

The WasteTwin project views the smart waste collection system as a SoS—a composition of multiple, independent, and distributed constituent systems (CSs) each possessing operational and managerial independence while contributing to a collective goal (Maier, 1998; Sage & Cuppan, 2001). The CSs concerned in this study are smart waste bins, waste collection trucks, and traffic lights.

Using a DT extend the capabilities of such systems by offering a virtual representation that synchronizes with its physical counterpart through continuous data exchange, allowing simulations, what-if scenarios, and predictive analytics (Tao et al., 2018). WasteTwin is based on the architecture in (Smati, Cheutet, Danjou, & Laval, 2025), but we simplified it to fit in the scope of the course. The realised SoS DT includes the following:

¹CityNews: <https://montreal.citynews.ca/2025/02/20/montreal-garbage-snow-removal/>

²Reddit: <https://shorturl.at/nF3jo>

³The Guardian: <https://shorturl.at/b6ybx>

⁴Ensemble Montréal: <https://ensemblemtl.org/en/actualites/cleanliness-issues-montreal/>

- a Digital Shadow of the trucks and traffic lights.
- Digital Twin of the bins that offers weather disturbance detection and visualization in real time using 3D assets created in Blender and integrated through the GoDot game engine, providing an immersive and responsive monitoring interface.

The full code of the described framework is available on GitHub⁵, as well as a video demonstrating its execution on the Drive⁶

The rest of this report is structured as follow:

⁵WasteTwin Code: <https://github.com/meriemsmt/wasteTwin>

⁶Demonstration: <https://drive.google.com/file/d/1ZRTbiGp1LT4oiMSsg7zpik4K-JCUwKmJ/view>

2 PHYSICAL TWIN DESCRIPTION AND ARCHITECTURE

The WasteTwin project is centred around the design and implementation of a DT for a smart waste collection system, conceptualized as a SoS. This SoS is represented in the form of a mock-up rather than real systems and it is composed of three autonomous yet interdependent CSs —smart waste bins, waste collection trucks, and urban traffic lights— each equipped with sensing or actuating capabilities and capable of operating independently, yet contributing to a shared city-wide objective: efficient, responsive, and resilient waste management.

This section presents an in-depth description of the architecture of Physical Twin (PT), from both a conceptual SoS perspective and a technical implementation viewpoint, covering the roles and interactions between CSs, the integration mechanisms and the data flow in the physical side.

2.1 Overview of the Constituent Systems

The WasteTwin SoS comprises three CSs, each with distinct roles and embedded intelligence. Figure .1 illustrates the used hardware and communication occurring between them, a detailed explanation is done below :

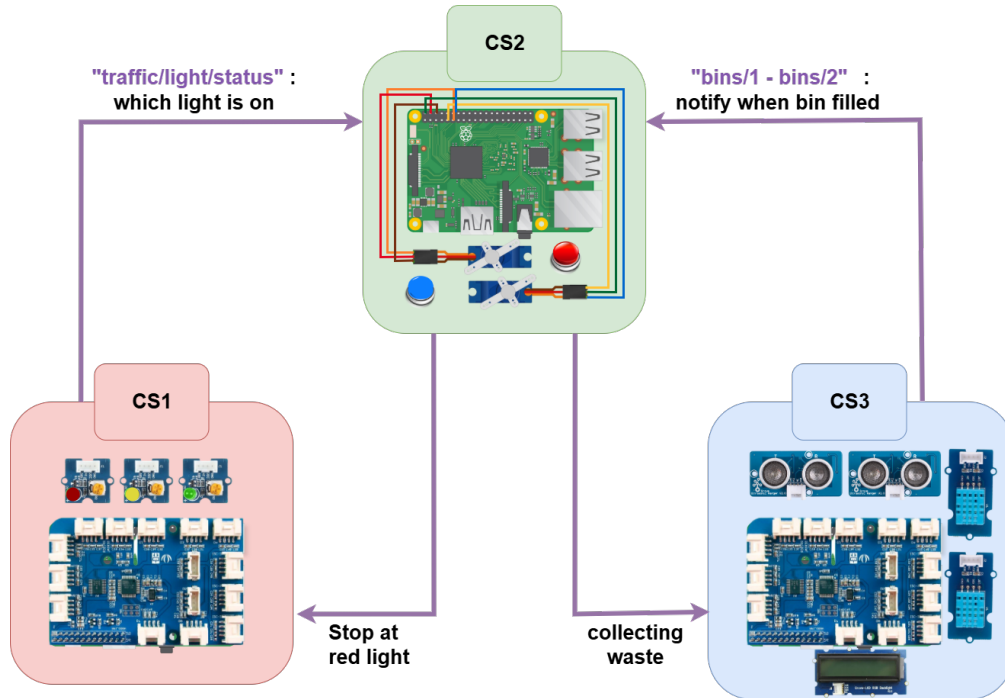


Figure .1 Physical Twin Hardware

- ***CS1 - Traffic Light System:*** The traffic light system serves as a supporting infrastructure, controlling vehicular flow and allowing or restricting access through intersections. In this prototype, the system is controlled independently by a Raspberry Pi, ie. showcasing the managerial independence, and the lights are modelled using LEDs of three different colours to mimic the real urban traffic lights in our mock-up. The operational independence is accomplished since this system has at a local operation which is switching the LEDs on and off depending on a delay. At the SoS level, CS1 communicates with the system of trucks (CS2) and affects indirectly the efficiency of waste collection operations by influencing truck travel time and accessibility. Its inclusion adds a layer of environmental complexity, illustrating emergent behaviour in the SoS.
- ***CS2 - Waste Collection Trucks:*** Two trucks are modelled using a Raspberry Pi, two micro servo motors and two push buttons. The buttons emulate the brakes of each truck so that when messages are received from the CS1, the trucks would stop by the red light. The trucks interact with the bins system (CS3) by updating their collection status and with traffic lights to optimize routing and avoid congested or disrupted zones.
- ***CS3 - Smart Waste Bins:*** Two bins are modelled using a Raspberry Pi and each equipped with an ultrasonic level sensor and environmental detectors (humidity and temperature sensors). Each bin collects in real-time data on their waste fill level and ambient conditions. Once the fill level exceeds a predefined threshold or an environmental event is detected (such as rain), the bin sends this data to CS2. The LCD actuator is used to close the loop coming from the DT and will be talked about in the next section.

Each CS operates with managerial and operational independence, is physically distributed, and evolves individually. When composed together in the WasteTwin framework, they demonstrate emergent properties, such as the ability to reconfigure operational plans in response to environmental changes or detect disruptions triggering a prioritization protocol. Figure .2 showcases the realized mock-up.

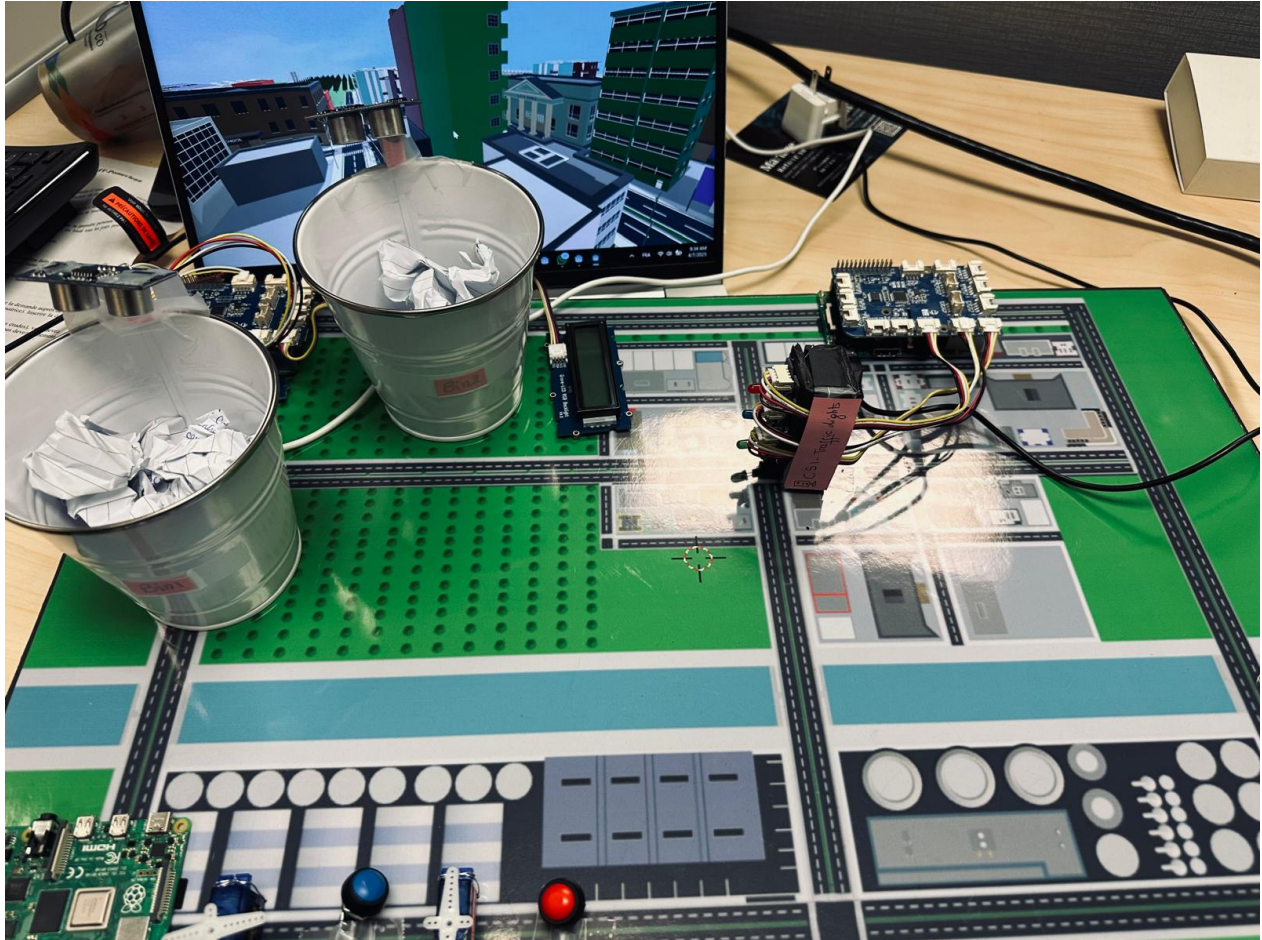


Figure .2 Physical Twin mock-up

3 DIGITAL TWIN ARCHITECTURE AND COMPONENTS

The realised DT plays a role in elevating the functionality, visibility, and responsiveness of the physical waste collection SoS. While the PT captures the operational essence of smart bins, trucks, and traffic lights, it is the DT that enables continuous observation, simulation and disturbance detection and real-time visualization.

This section presents a comprehensive overview of the DT’s architecture, which was designed with modularity, interoperability, and responsiveness in mind. It discusses how the DT maps to the physical systems, the enablers that facilitate communication and synchronization, and the layers of functionality that empower the digital ecosystem. Figures .3 and .4 provide complementary visual explanations of the DT system’s structure and behaviour.

3.1 DT Overview

Since the DT needs to replicate with the maximum of fidelity possible the PT, so the characteristics of the SoS (managerial and operational independence) needs to be applied on the digital side as well.

As shown in Figure .3, the SoS CSs have either a DT or a Digital Shadow (DS) where:

- DS1 mirrors the traffic lights system.
- DS2 mirrors the fleet of collection trucks.
- DS3 mirrors the smart waste bins.

The modular design enables each DS/DT to operate semi-independently (only depending on its physical counterpart), while data and commands are exchanged through topic-based messaging systems using **MQTT** and **RabbitMQ**⁷. MQTT handles low-latency, lightweight telemetry between physical sensors and digital replicas, while RabbitMQ ensures structured messaging, particularly for actions requiring reliability (e.g., notification delivery, collection commands).

3.2 DT Constellation and Services

The Digital Twin Constellation is visualized in Figure .4 and follows the architecture proposed in (Oakes et al., 2024). It presents the full mapping between physical and digital elements, including the enablers, services, and communication flows required for each system component.

⁷<https://www.rabbitmq.com/docs/mqtt>

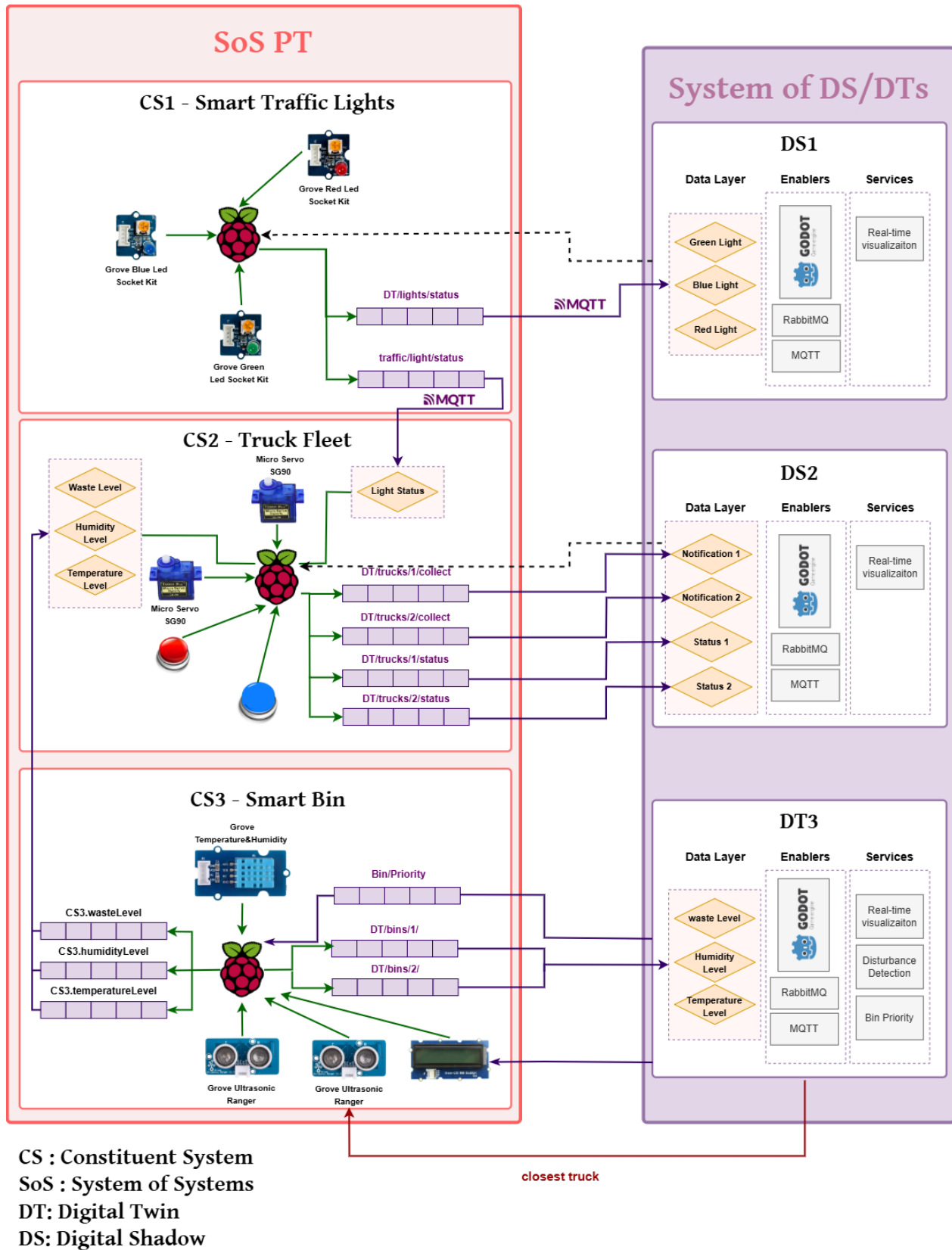


Figure .3 Architecture of the SoS and its DT

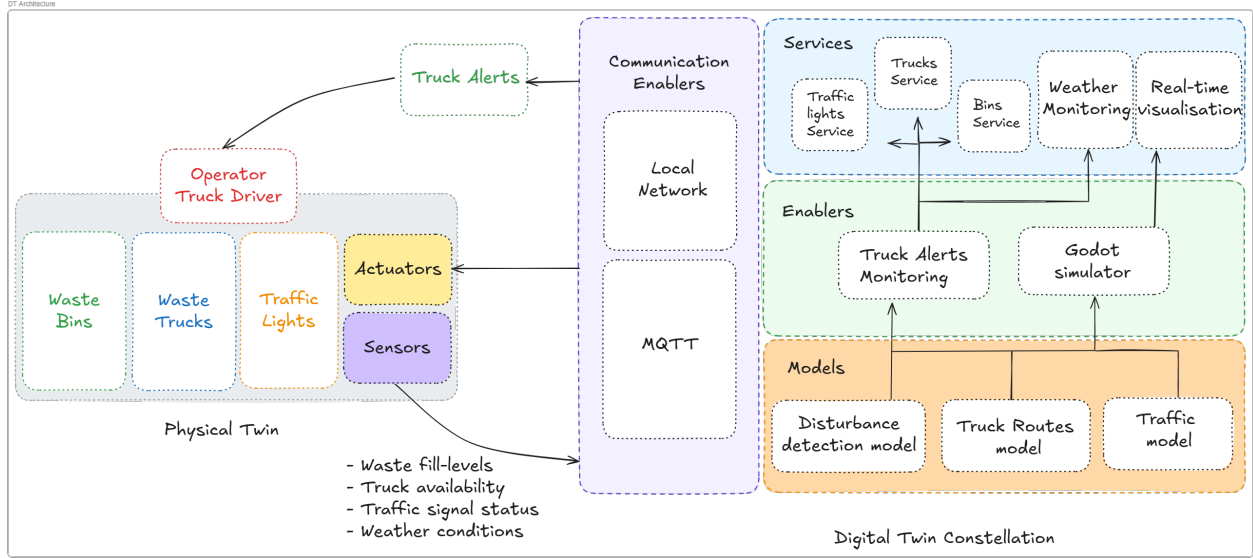


Figure .4 SoS and DT constellation

3.2.1 PT

On the left side of the architecture lie the CSs: smart bins, trucks, and traffic lights, each equipped with sensors and actuators. These devices continuously emit telemetry data—such as:

- Waste fill level and environmental variables (humidity, temperature) from bins,
- Location status and brake states from trucks,
- Signal states from LEDs in the traffic lights.

These independent CSs communicate over a local network and the collected data is passed through MQTT channels into the DT environment. The DT subscribes to these topics and updates its internal models and 3D representation accordingly.

3.2.2 Communication Enablers

The communication layer consists of a local network and two key technologies:

- MQTT for data telemetry and command publishing.
- RabbitMQ for reliable task queuing and asynchronous message routing which is important in an architecture based on SoSs.

These enablers guarantee interoperability across heterogeneous devices and ensure that delays or failures in one subsystem do not cascade into others—a crucial property of resilient systems.

3.2.3 Offered Services

Once data is ingested and well received in the DT, the DT system offers several services which are the following:

- **Truck Routing Service:** Identifies the optimal truck to dispatch based on bin priority, current location, and environmental conditions.
- **Traffic Lights Service:** Displays current light signals and allows visual feedback on accessibility.
- **Bin Service:** Detects bin overflow, assigns collection priority, and provides feedback to the physical LCD display on the bin.
- **Weather Monitoring Service:** Integrates environmental sensors to simulate disturbance conditions such as rain, which may delay collection operations or trigger re-routing.
- **Real-Time Visualization:** The Godot engine displays the city layout, dynamic bin states (color-coded), truck movement, and traffic status.

These services are supported by a modelling layer which includes:

- A Disturbance Detection Model for simulating rain or service interruption,
- A Truck Routing Model using shortest-distance heuristics,
- A Traffic Simulation Model that reflects light behaviour and its effect on truck mobility.

3.2.4 Bidirectional Feedback and Adaptive Behaviour

As any DT, the DT of CS3 has a closed-loop feedback structure. It does not operate as a passive observer like the DSs; it actively reconfigures operations in response to state changes. For example:

- When a bin reaches 80% fill and rain is detected, the DT calculates which truck is closest and least affected by traffic delays and sends a collect command to the corresponding truck's topic.
- Simultaneously, it updates the digital scene to reflect bin overflow and impending collection.
- If a truck is delayed or blocked by a red signal, the DT postpones dispatch or reroutes another available truck, ensuring continuous operation without human intervention.

This real-time decision loop demonstrates resilience through self-organization, a property emphasized in SoS engineering and resilience theory (Hosseini et al., 2016). The system avoids collapse under stress and maintains service quality by adapting dynamically to internal and external disturbances.

4 Reporting Characteristics

Table .1 Reporting characteristics of the WasteTwin Digital Twin system (based on Gil et al., 2024).

ID	Characteristic	WasteTwin Implementation
MC1	System under Study	Smart waste collection System of Systems composed of smart bins, collection trucks, and traffic lights, implemented as a physical mock-up.
MC2	Physical Acting Components	The DT acts on the LCD of the Bins to inform the truck operator that there is a disturbance.
MC3	Physical Sensing Components	Ultrasonic sensors (fill levels), LEDs status, Micro Servo motors status, DHT sensors (temperature, humidity), and push buttons (truck brakes).
MC4	Physical-to-Virtual Interaction	Sensor data from lights, bins and trucks is sent to the DT using MQTT topics.
MC5	Virtual-to-Physical Interaction	Updating the LCD display and identifying the closest truck are sent to CS3 using MQTT and RabbitMQ.
MC6	Digital Twin Services	Real-time 3D visualization, environmental disturbance detection, disturbance detection, bin prioritization.
MC7	Twinning Time-scale	Real-time synchronization: events and updates reflected immediately in the digital environment.
MC8	Multiplicities	One-to-one mapping: each physical system has a corresponding digital subsystem (DS1, DS2, DT3).
MC9	Lifecycle Stages	Operation phase.

ID	Characteristic	WasteTwin Implementation
MC10	DT Models and Data	Rule-based models for bin prioritization, truck dispatch logic, weather influence; 3D spatial models in Godot.
MC11	Tooling and Enablers	Blender, Godot, GDScript, MQTT, RabbitMQ, and local networking.
MC12	DT Constellation	Three synchronized digital subsystems (DS1: traffic, DS2: trucks, DT3: bins), communicating via a shared messaging layer.
MC13	Twinning Process and Evolution	Developed iteratively for the LOG6953FE course, a simplified architecture from a previous contribution ((Smati et al., 2025)).
MC14	Fidelity and Validity Considerations	Accuracy checked through real-time visual mirroring of states; physical responses match digital triggers.
MC15	DT Technical Connection	MQTT for lightweight sensor updates; RabbitMQ for control messaging and feedback routing.
MC16	DT Hosting / Deployment	DT hosted locally on a PC; each PT runs independently on distributed Raspberry Pis.
MC17	Insights and Decision Making	Visual insights into waste status, delays, bin overflow, and weather effects support human evaluation.
MC18	Horizontal Integration	Potential for expansion toward other smart systems (e.g., energy, mobility), currently focused on internal integration.
MC19	<i>Data Ownership and Privacy</i>	<i>Academic prototype; data remains locally stored and not shared externally. No personal data collected for now.</i>
MC20	<i>Standardization</i>	<i>following the standards of SoS Engineering and the architecture of (Smati et al., 2025).</i>
MC21	<i>Security and Safety Considerations</i>	<i>No external access; local-only system ensures safe testing environment without cybersecurity exposure.</i>

5 Contributions and met Problems

The realised Gantt Diagram presented in Figure .5 showcases the timeline with approximate dates and the contribution of both students.

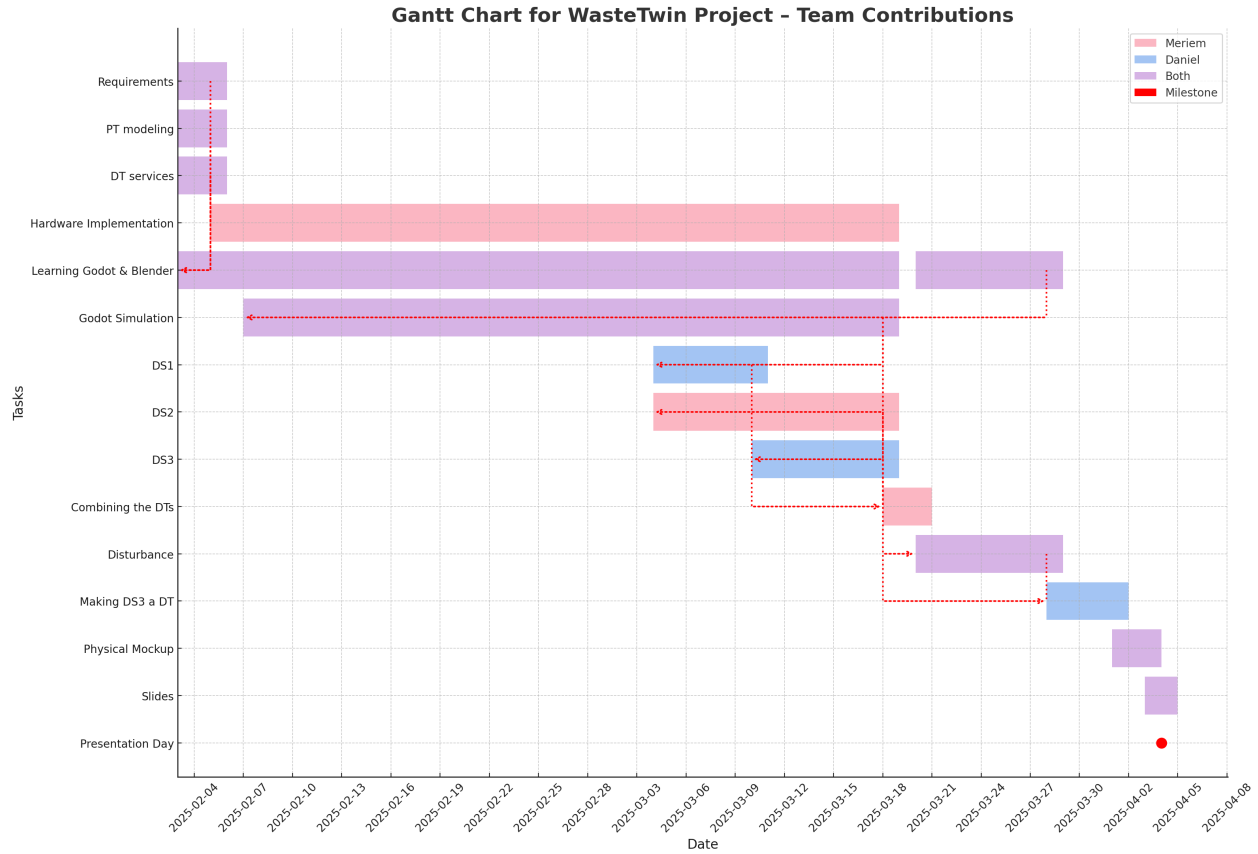


Figure .5 Gantt Diagram

The realization of this project involved several practical challenges, including:

- **Power supply issues:** Raspberry Pis would shut down unexpectedly when not adequately powered, occasionally resulting in sensor damage and loss of unsaved code.
- **Tool instability:** Godot, the engine used for 3D visualization, undergoes frequent updates with significant changes between versions, making it difficult to find up-to-date tutorials or documentation relevant to the version used in this project.

6 Conclusion

The WasteTwin project presented in this report demonstrates a functional use of a DT designed to support the operation and resilience of a smart waste collection SoS. By integrating physical components—including smart bins, waste collection trucks, and traffic lights—with a digital simulation environment, WasteTwin achieves real-time synchronization between the physical and virtual layers, enabling dynamic monitoring, event-driven responses, and immersive visualization.

Throughout the development of this system, key principles of DT architecture, IoT-based communication, and SoS engineering have been put into practice. The project illustrates how DTs can enhance the operational efficiency of waste management systems by detecting disruptions (e.g., rain, overflow, blocked routes), prioritizing interventions, and adapting routing strategies in real time. The modular SoS architecture enables independent control and monitoring of each subsystem, while the use of lightweight messaging protocols (MQTT, RabbitMQ) facilitates low-latency communication across all layers of the system.

Despite encountering practical challenges—such as hardware sensitivity and tooling limitations—the project successfully demonstrates the feasibility of creating a responsive, interactive DT prototype within the constraints of an academic environment.

In conclusion, WasteTwin offers a compelling example of how DTs can transform static urban operations into dynamic, data-driven systems. Future work may include incorporating predictive analytics, extending horizontal integration with other systems (e.g., energy, mobility), and exploring standardized frameworks for industrial interoperability. As smart cities continue to evolve, the use of DTs such as WasteTwin will become increasingly central to achieving resilient, efficient, and sustainable urban management.

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